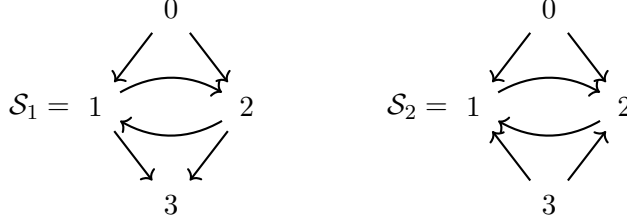


For any simplicial complex Σ we can consider its face poset $F(\Sigma)$, whose nerve is (the barycentric subdivision of) Σ . Then, $\text{Inj}(F(\Sigma))$ is homotopy equivalent to Σ , and the complex of injective words on directed graphs has any homotopy type.

Example 4.8. Consider the following digraphs with reciprocal edges:



Note that the associated complexes of injective words are isomorphic (and homeomorphic to the 2-sphere). By Theorem 3.9, the graph S_2 is regularly diagonal. Also the graph S_1 is regularly diagonal. However, the associated eulerian magnitude homology groups are not isomorphic, such as in bidegree $(3, 3)$.

The last example shows that, although the complexes of injective words on two graphs are homotopy equivalent, we can have different eulerian magnitude homology groups. In the next section we aim to investigate this property more accurately.

5. THE INJECTIVE NERVE OF CATEGORIES AND THE REGULAR MAGNITUDE-PATH SPECTRAL SEQUENCE

The goal of this section is construct the regular magnitude-path spectral sequence, in analogy with the magnitude-path spectral sequence developed in [Asa23]. To do so, it shall be more convenient to use the categorical language, and to generalize the framework of eulerian magnitude homology to quivers. In the first subsection we introduce the tool of *injective nerves* of categories, and then we shall apply it to the reachable categories, getting the regular magnitude-path spectral sequence. As an application of the construction, we shall provide some computations of regular path homology of regularly diagonal graphs.

5.1. The injective nerve. Let $\mathbf{2}$ denote the category consisting of the objects E and V , with two non-identity morphisms $s, t: E \rightarrow V$ called the *source* and the *target*. We shall also represent, and identify, a quiver Q by the corresponding directed graph with set of vertices V , and set of edges E . Let $\mathbf{Fin}$ be the full subcategory of $\mathbf{Set}$ of finite sets. By a (finite) *quiver* we shall mean a functor $Q: \mathbf{2} \rightarrow \mathbf{Fin}$. Morphisms of quivers are natural transformations of functors. The category **Quiver** of finite quivers and morphisms of quivers is the functor category $\mathbf{Fun}(\mathbf{2}, \mathbf{Fin})$.

There is an adjunction

$$(6) \quad \begin{array}{ccc} & \xrightarrow{\text{Free}} & \\ \mathbf{Quiver} & & \mathbf{Cat} \\ & \xleftarrow{U} & \end{array}$$

between the category of quivers and the 1-category **Cat** of small categories and functors. In this adjunction, the functor *Free* is the free functor that turns a quiver into a category, and *U* is the corresponding forgetful functor.

Recall that an *ordered k -clique* of a directed graph G is a totally ordered k -tuple $(v_1, \dots, v_k)$ of vertices of G with the property that, for every $i < j$, the pair (v_i, v_j) is an ordered edge of G . This definition extends to quivers in the natural way and, for any given quiver Q , we can construct the associated directed flag complex $\text{dFl}(Q)$ as